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Drilling the World's Longest Horizontal Well Using an Environmentally Friendly System Utilizing State-of-the-Art Water-Based Drilling Fluid for Ultra-Extended Reach Drilling (ERD)

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Abstract

The deployment of advanced water-based drilling fluid technology facilitated the successful completion of the world's longest horizontal well lateral, extending 18,508 feet in a demanding high-temperature, sour environment. This achievement highlights significant advancements in environmental sustainability, cost efficiency, and technical innovation within ultra-extended reach drilling (ERD). In response to increasing demands for sustainable and economical drilling solutions, this paper details the execution of a record-setting 6" horizontal lateral, measuring 18,508 feet, utilizing a water-based mud (WBM) system for an ERD application. The well, drilled in a congested onshore field, achieved a total measured depth of 30,223 feet, establishing a new global benchmark for horizontal reach with WBM. The operation prioritized environmental stewardship, reservoir compatibility, and economic efficiency by replacing conventional oil/synthetic-based muds with a non-damaging WBM engineered from calcium chloride brine, thermally stable polymers, and a marble-based bridging package. This formulation enabled the creation of a thin, impermeable filter cake, minimizing formation damage and eliminating the need for breaker fluids. To address high bottomhole temperatures exceeding 300°F (149°C), dual mud chillers and a premium lubricant were deployed to maintain fluid integrity and reduce torque and drag. Laboratory simulations and contingency planning ensured fluid stability and tool performance across tripping and completion phases. The operation demonstrated exceptional fluid stability for over 144 hours, zero fluid-induced formation damage, optimized rheology for hole cleaning, and significant cost savings compared to traditional non-aqueous fluid systems.

This achievement redefines the technical limits of WBM in ERD, extending its applicability beyond the historical 14,000-foot lateral threshold. It sets a new industry standard for environmentally conscious drilling, offering a viable alternative for unlocking complex reservoirs with reduced surface infrastructure and emissions.

Keywords: Drilling the world's longest Horizontal well, Water-based drilling fluids for Ultra-extended reach, cost effectiveness, minimizing formation damaged

Introduction

ERD is an operative practice for drilling high-angle wellbores with long horizontal displacement. Typically, a well design is classified as ERD when the horizontal displacement is at least two times (Angelo, Shabaka, Hamed, Vinci, & Abd El Salam, 26-28, 2016)) the vertical displacement true vertical depth (TVD). The longest well drilled in the UAE reached a measured depth of 30,223 feet, including an 18,508-foot horizontal lateral (Figure 1). While technically challenging and more expensive initially, this method offers greater reservoir contact and faster returns, making it the industry's preferred approach over conventional designs.

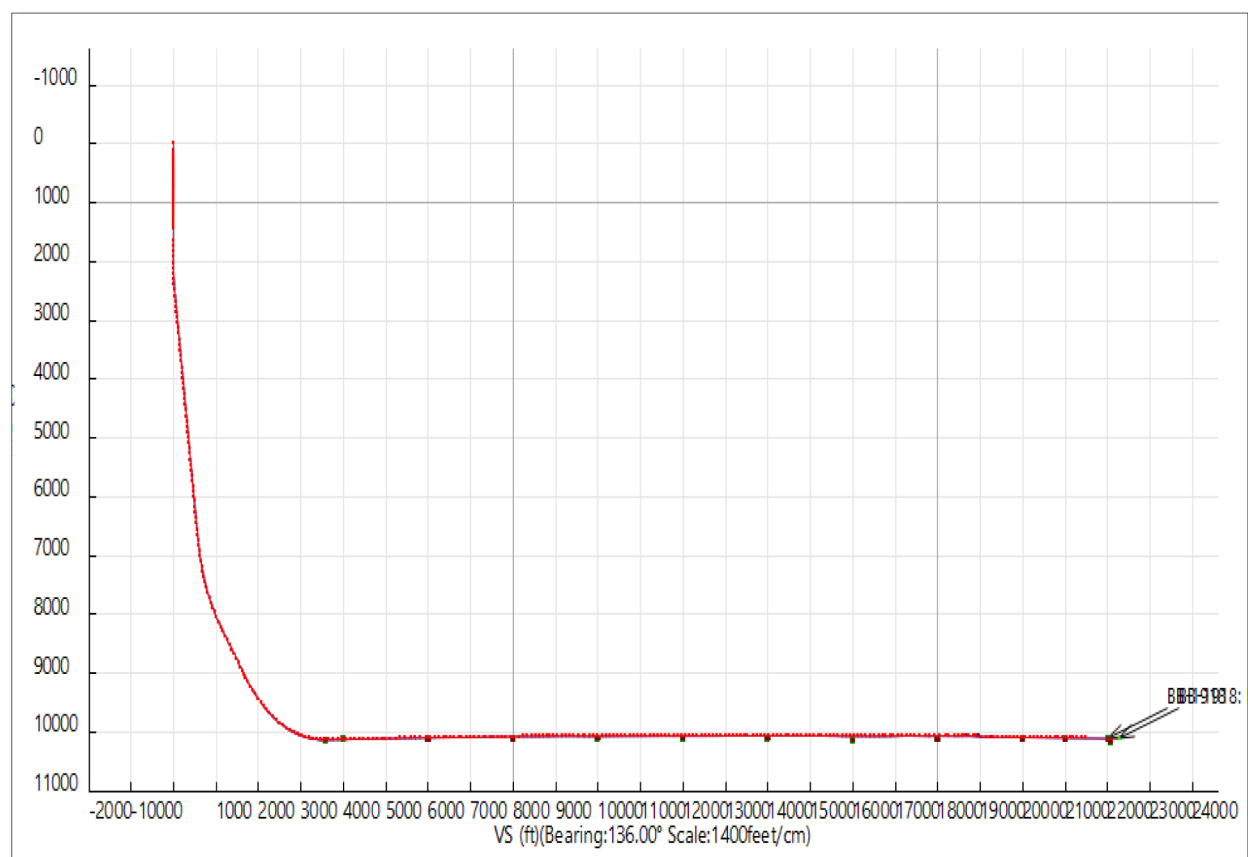


Figure 1—Well profile

Oil Field background

In our case studies, the hydrocarbon type is light oil, and the typical formation lithology is limestone. Production wells are generally highly deviated and horizontal. The well position on the one of the most prolific onshore oil fields, presents a unique set of geological and operational challenges due to its complex carbonate reservoir system, high temperature gradients, and naturally fractured formations. As part of 's continuous efforts to maximize reservoir contact and enhance hydrocarbon recovery, the longest well was planned and executed to drill the longest horizontal reservoir section in the field's history. The well was drilled through a highly heterogeneous carbonate formation characterized by depleted zones, variable pore pressure, and faulted structures. The operation required meticulous geosteering, advanced drilling fluids engineering, and optimized bottom hole assembly (BHA) design to ensure wellbore stability, minimize non-productive time (NPT), and achieve precise reservoir placement. Key innovations included the use of a high-

performance water-based mud system tailored to withstand extreme downhole conditions—temperatures exceeding 295°F. The drilling fluids program was critical in mitigating differential sticking, controlling fluid loss, and maintaining wellbore integrity throughout the extended reach section. One of the concerns includes the importance of wellbore strengthening techniques, real-time data integration, and collaborative planning across drilling, fluids, and subsurface teams. The successful execution of this well sets a benchmark for future extended reach drilling (ERD) projects in mature carbonate reservoirs across the region ([Cameron, October 2001](#)).

Preliminary Laboratory Studies

The primary objectives of the CaCl₂ WBM Formulation design for this well were designed considering the following:

- Reduce the ECD by reducing the amount of solids in the fluid.
- Achieve tight fluid loss.
- Maintain low sag factor because of longer liner running times, with the fluid being static for a prolonged time.

A systematic testing process was used to develop a novel solution for well challenges, with key focus on torque reduction, wellbore stability, stable rheology, and fragile gel strength. Only ground marble was allowed as the reservoir fluid's weighting and bridging agent, aiming for minimal solids concentration. These considerations led to a tailored drilling fluid that improved lubricity, hole stability, trajectory control, contamination resistance, and hydrocarbon yield ([Payne, Cocking, & Hatch, September 25-28, 1994](#))

To achieve these criteria, a customized fluid system formulation was designed using the following methodology, Calcium chloride brine was selected as a representative di-valent brine system. The lubricant concentrations tested were 1%, 2%, and 3%. Additional parameters observed during testing included physical separation of the lubricant, cheesing effect, and greasing effect on both fresh and heat-aged fluids.

Testing Procedure

- The testing protocol included the following steps:
- Preparation for CaCl₂ brine mud.
- [Table 1](#) - Shows CaCl₂ WBM Formulation.
- Evaluation of lubricity coefficient and torque for all formulated brines.

Table 1—Shows CaCl₂ WBM Formulation

	Blank	Formulation#1	Formulation#2	Formulation#3
		Lubricant 1%	Lubricant 2%	Lubricant 3%
Product	lb/bbl	lb/bbl	lb/bbl	lb/bbl
CaCl ₂ Brine	403.79	399.52	395.24	390.96
VISCOSIFIER	1.20	1.20	1.20	1.20
HIGH TEMPERATURE FLUID LOSS REDUCERS	6.00	6.00	6.00	6.00
STARCH	4.00	4.00	4.00	4.00
BUFFER	1.00	1.00	1.00	1.00
MARBLE	30.00	30.00	30.00	30.00
H ₂ S SCAVENGER	0.25	0.25	0.25	0.25
BIOCIDE	0.25	0.25	0.25	0.25
OXYGEN SCAVENGER	0.15	0.15	0.15	0.15
LUBRICANT		3.51	7.01	10.52

The equipment used for lubricity testing was as follows:

- EP Lubricity Tester
- Type of Lubricity Evaluated: Metal to Metal Lubricity
- Pressure Applied During Testing: 150 PSI, 200 PSI, and 300 PSI

The test method for evaluating lubricity is outlined below, referencing the SOP for Lubricity Test.

The procedure consists of two stages

Stage 1 - Calibration of the Lubricity Tester

- Clean the ring, block, and holder with acetone.
- Submerge ring and block in fresh deionized water.
- Set instrument speed to 60 rpm; zero torque and timer.
- Apply 150 in-lb of force for five minutes; allow reading to stabilize.
- Repeat process three times.
- If measurements are not within 34 ± 2 , further grind block and ring:
- For readings <32 : use coarse compound, 1 hour at 100 rpm/90 in-lb, redistribute every 15-30 min.
- For readings >38 : use fine compound, 1 hour at 100 rpm/75 in-lb, redistribute as need-ed.
- Adjust until results fall within specified range, then proceed with analysis.

Stage 2 - Evaluation of Lubricity Coefficient and Torque for Different Fluids

- Only conduct the lubricity test after calibrating the unit with deionized water and achieving a coefficient of friction of 34 ± 2 .

- Fill the stainless-steel sample cup with 260-280 mL of test fluid and position it correctly.
- Raise the cup stand to submerge components, secure the cup, and zero the torque reading.
- Ensure the torque arm and test block are in place; adjust the handle to a gauge reading of 150 inch-pounds.
- Apply torque only when the test ring is submerged.
- Zero the timer, run the machine for three minutes, and record the torque reading.
- Release the torque, reset motor speed, lower the cup stand, discard fluid, and thoroughly clean all parts.

The multiple formulations were tested. [Table 2](#) - show CaCl₂ WBM drilling fluid testing results.

Table 2—show CaCl₂ WBM drilling fluid testing results.

		Formulation Blank		Formulation#1 Lubricant 1%		Formulation#2 Lubricant 2%		Formulation#3 Lubricant 3%	
Formulation description:		BHR	AHR	BHR	AHR	BHR	AHR	BHR	AHR
	HOURS		16		16		16		16
Pressure	PSI		100		100		100		100
Temp	°F		250		250		250		250
RHEOLOGY 20 °F	600 RPM	65	56	56	48	55	47	60	52
	300 RPM	41	35	32	27	34	29	37	32
	200 RPM	29	26	22	20	25	22	27	24
	100 RPM	17	15	11	10	15	13	16	14
	6 RPM	2	1	2	1	2	1	2	1
	3 RPM	1	0.5	1	0.5	1	0.6	1	0.5
	PV CP	24	21	24	21	21	18	23	20
	Y P LBS/100FT ²	17	14	8	6	13	11	14	12
Lubricity	TORQUE READING 150 PSI @ RT		14.9		12.7		11.5		10.6
	LUBRICITY COEFFICIENT 150 PSI @ RT		1.152		0.129		0.117		0.18
	% TORQUE REDUCTION				14.8		22.8		28.9

Testing Conclusions

After testing completed Lubricant shown excellent performance after the hot rolling in CaCl₂ WBM showing high torque reduction percentage and Lubricity coefficient specially in the higher concentration of 2% and 3% by volume and proving that it's compatible with the CaCl₂ based fluids ([Figure 2](#) - Torque reduction percentage and Lubricity coefficient results).

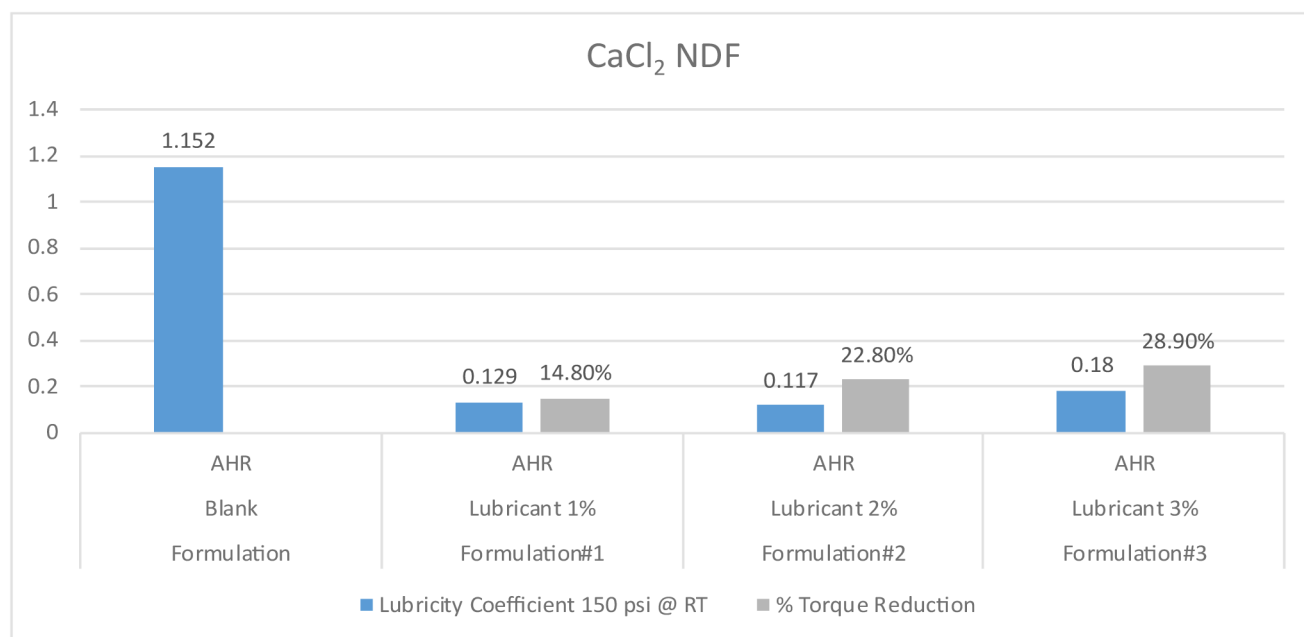


Figure 2—Torque reduction percentage and Lubricity coefficient results

Execution and Field Results

A CaCl₂ WBM fluid system was previously implemented in the field, demonstrating excellent performance and meeting all critical criteria required for drilling extended lateral sections. However, exceptionally long intervals—such as an 18,508 ft section—posed additional operational challenges. While laboratory testing and initial field applications confirmed the system's high reliability, increased footage introduced complexities related to ECD, torque management, tool capability, fluid stability over prolonged periods, and logistics.

Challenges

Drill the 6" horizontal reservoir section to total depth 30,223 ft MD, ensuring optimal hydraulic performance, minimizing formation damage, and achieving the desired friction-factor parameters while addressing high torque and friction challenges.

- Implement fluid treatment and dilution strategies to address carbonate and bicarbonate contamination.
- Optimize bridging agent selection and application.
- Enhance loss control measures.
- Maintain high-temperature fluid stability during extended tripping operations.
- Mitigate thermal degradation effects.

Preparation

- Agreed and established fluid property targets with Client specs.
- Fluid service provider delivers premium high temperature lubricants that reduce friction factors while maintaining performance under extreme downhole conditions.

Job Execution

The NDF Reservoir Drilling Fluid: engineered Non-Damaging Water-Based Mud (WBM) system utilizes an environmentally benign divalent (Calcium Chloride) brine coupled with a precise formulation of non-damaging polymers and a bridging package. Components synergistically create an impermeable, thin filter cake that minimizes formation damage, ensuring negligible skin effect and preserving reservoir integrity. This stable fluid system maintained its integrity under the prescribed bottom-hole conditions, demonstrating tolerance to contamination and efficacy under static conditions, which were essential for executing the targeted well activities.

Implemented rigorous treatment and dilution protocols for the NDF fluid to effectively mitigate carbonate and bicarbonate contamination, ensuring fluid stability and performance. Applied a tailored blend of fine and medium calcium carbonate to enhance wellbore sealing, significantly reducing losses—particularly when drilling through fault zones.

Rheology Profile

PV, YP, LSYP, low end readings, and Tau 0 are important parameters in the execution of ERD wells. For effective hole cleanup, these values should be planned and maintained throughout the section. During this section, maintaining a minimal PV (to limit its impact on the ECD) and a Tau 0 greater than 5 lbm/100 ft² was required to support adequate hole cleaning and manage cuttings load, which in turn affected the ECD value (Mahrous, et al., 22 - 24 August 2016). Low end readings were maintained for hole cleaning efficiency without adverse effects on the rheology profile. All these parameters were properly managed by adding a novel polymers package that allowed for rheological profile modification as dictated by the well. Figure 3 - showing the rheological profile while drilling.

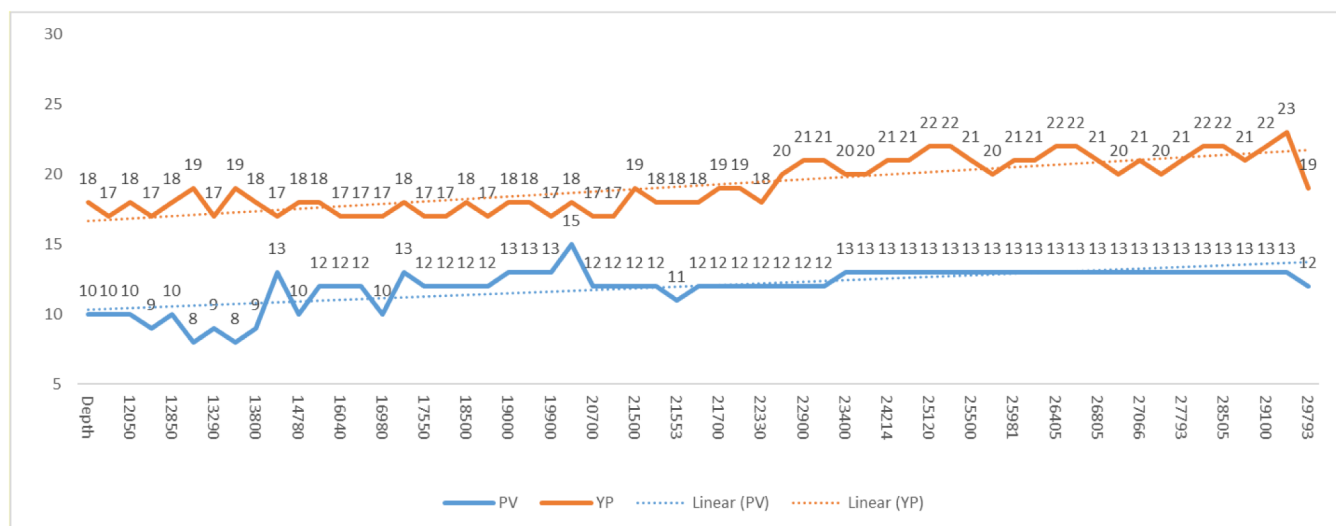


Figure 3—showing the rheological profile while drilling

Low-end readings were consistently maintained to optimize hole cleaning efficiency, with no negative impact on the rheology profile. These parameters were effectively managed through the introduction of an innovative polymer package, enabling targeted modification of the rheology profile in accordance with well requirements. Figure 4 - showing the Low-end readings profile while drilling.

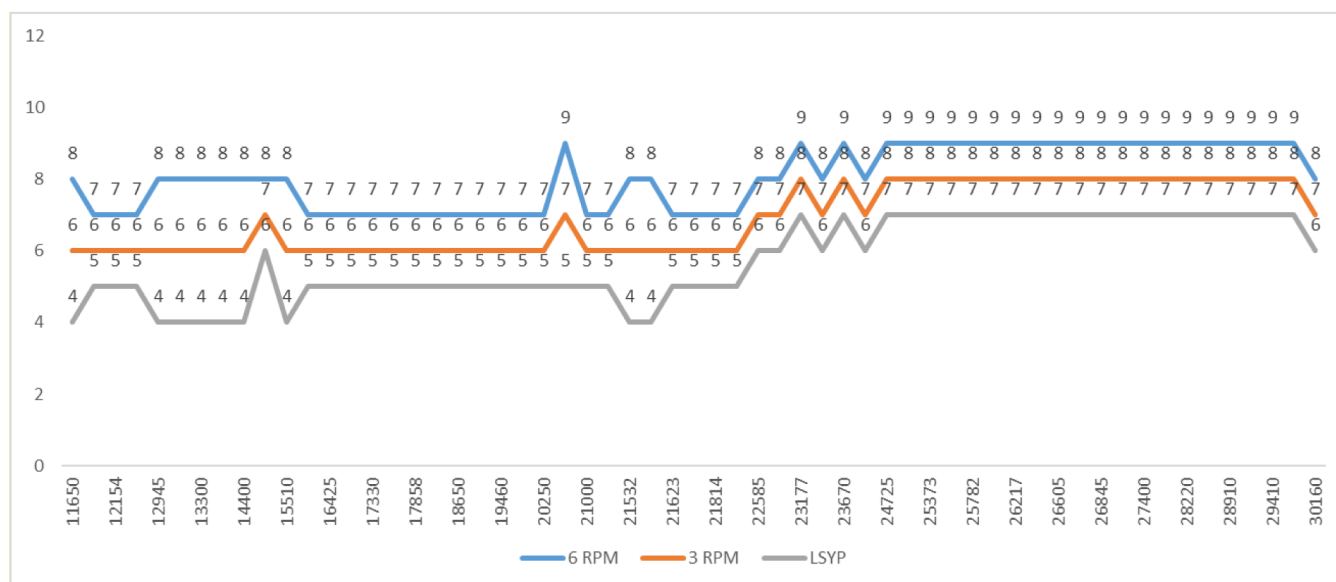


Figure 4—showing the Low-end readings profile while drilling

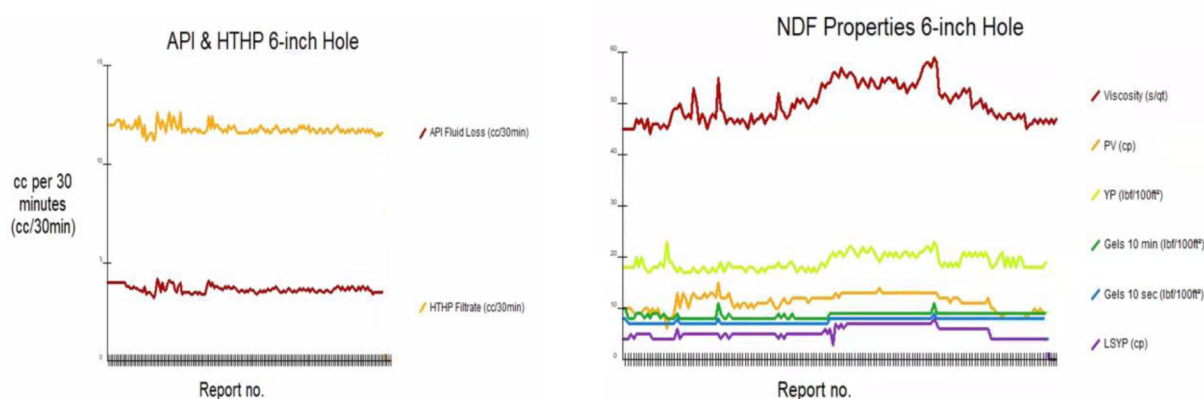


Figure 5—showing fluid loss & rheological profile while drilling

Friction Factor (FF): Torque and Drag

The friction factor is a critical parameter in ensuring the successful deployment of tubulars to total depth (TD) without complications (Abdel Salam, Goswami, & Hamed, March 22-25, 2016). Deploying 30,223 ft of tubulars—of which over 18,508 ft are situated on the low side of the well—results in significant metal-to-formation friction. The NDF Reservoir Drilling Fluid should provide proper lubricity to minimize these FFs. During operations, it was evident the NDF Reservoir Drilling Fluid system provided a low FF and mitigated stress issues in the tubulars. Figure 6 - showing the reduction of friction factor by the addition of lubricant while drilling.

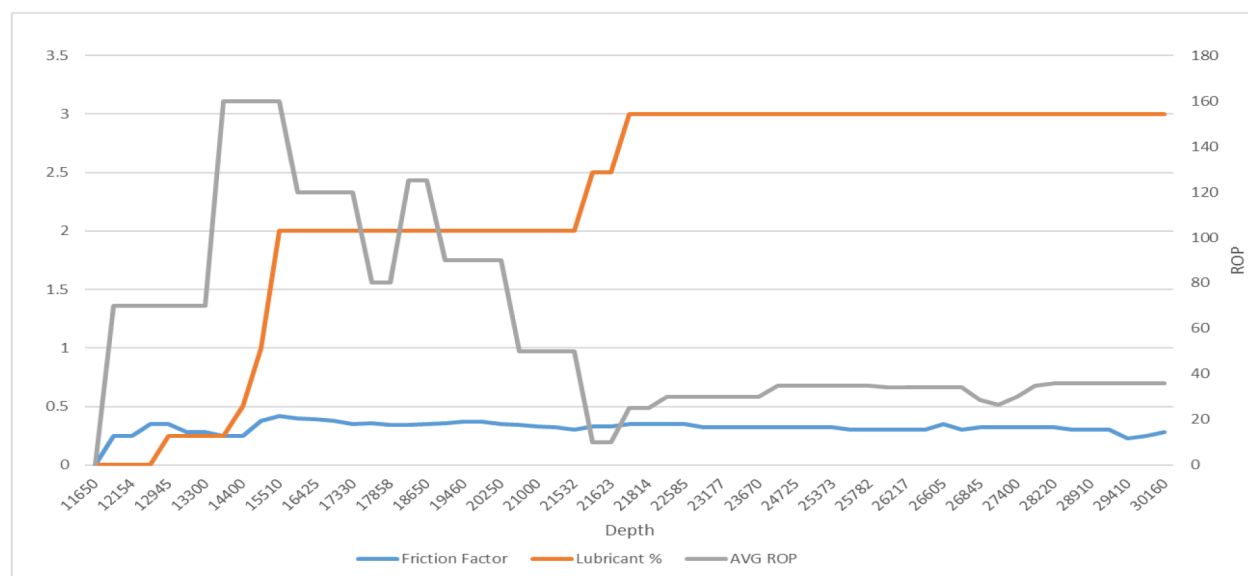


Figure 6—showing the reduction of friction factor by the addition of lubricant while drilling

Torque and drag data was recorded during section execution, these data shown the performance of the NDF Reservoir Drilling Fluid without exceeding torque and drag values.

Optimized fluid formulation with high-temperature stabilizers and premium lubricant to prevent rheological breakdown and control elevated torque and drag. Only 3% v/v of premium lubricant was required to significantly reduce the friction factor. The mud rheology was carefully optimized to reduce ECD and pump pressure while maintaining optimal hole cleaning efficiency. Successfully maintained drilling fluid integrity while exposed to 290°F (143°C) downhole temperatures for 144-192 hours during critical tripping operations.

Hole Cleaning and ECD

Hole cleaning received particular focus due to the planned long lateral section (>18,000 ft). Optimizing low-end rheology was essential to achieve the desired profile and maximize hole cleanliness. The drilling fluids provider employed the Herschel-Bulkley model to assess cleaning efficiency, maintaining Tau Zero above 5 lb/100 ft² throughout drilling, which ensured stable mud properties without overtreatment.

To optimize operations, ECD profiles were predicted using the planned well trajectory and possible parameter setups, giving useful guidance during drilling. Figure 8 ECD, Hole cleaning during drilling. Figure 9. ECD, Hole cleaning during drilling. Figures 7 and 8 illustrate software simulations of cuttings load, transport efficiency, average velocity, and ECD values with and without cuttings.

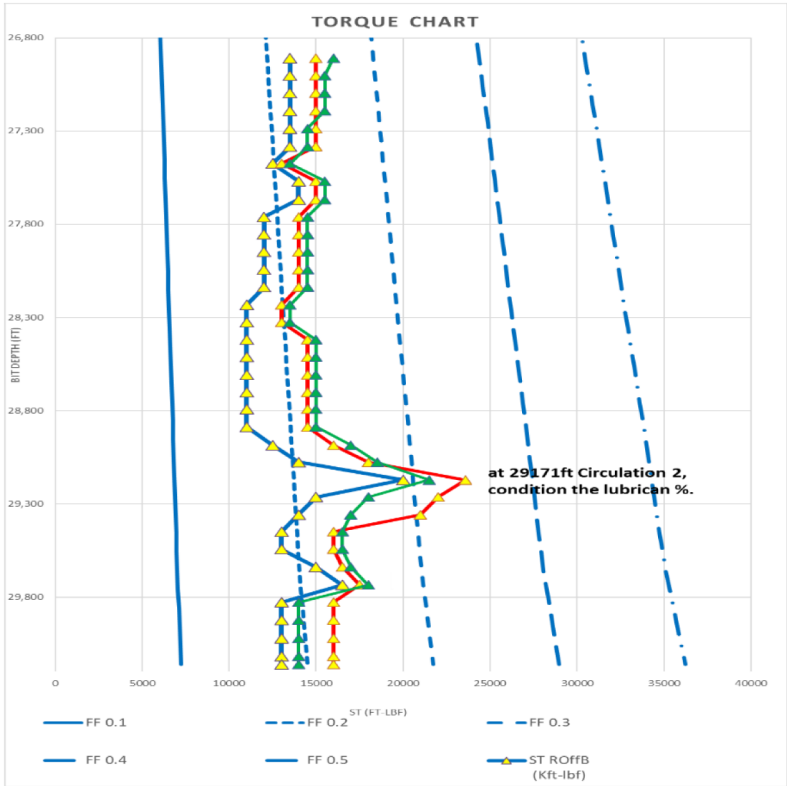


Figure 7—shown the torque reading while drilling

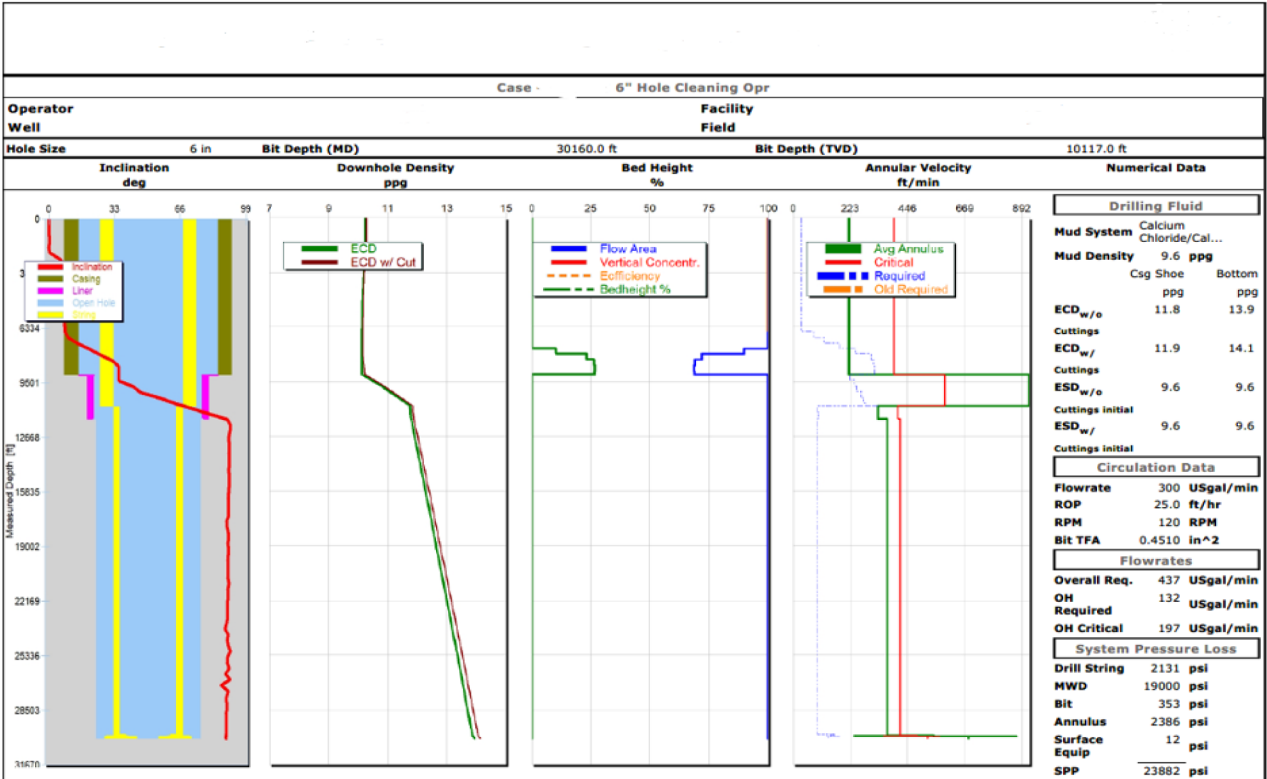


Figure 8—ECD, Hole cleaning during drilling.

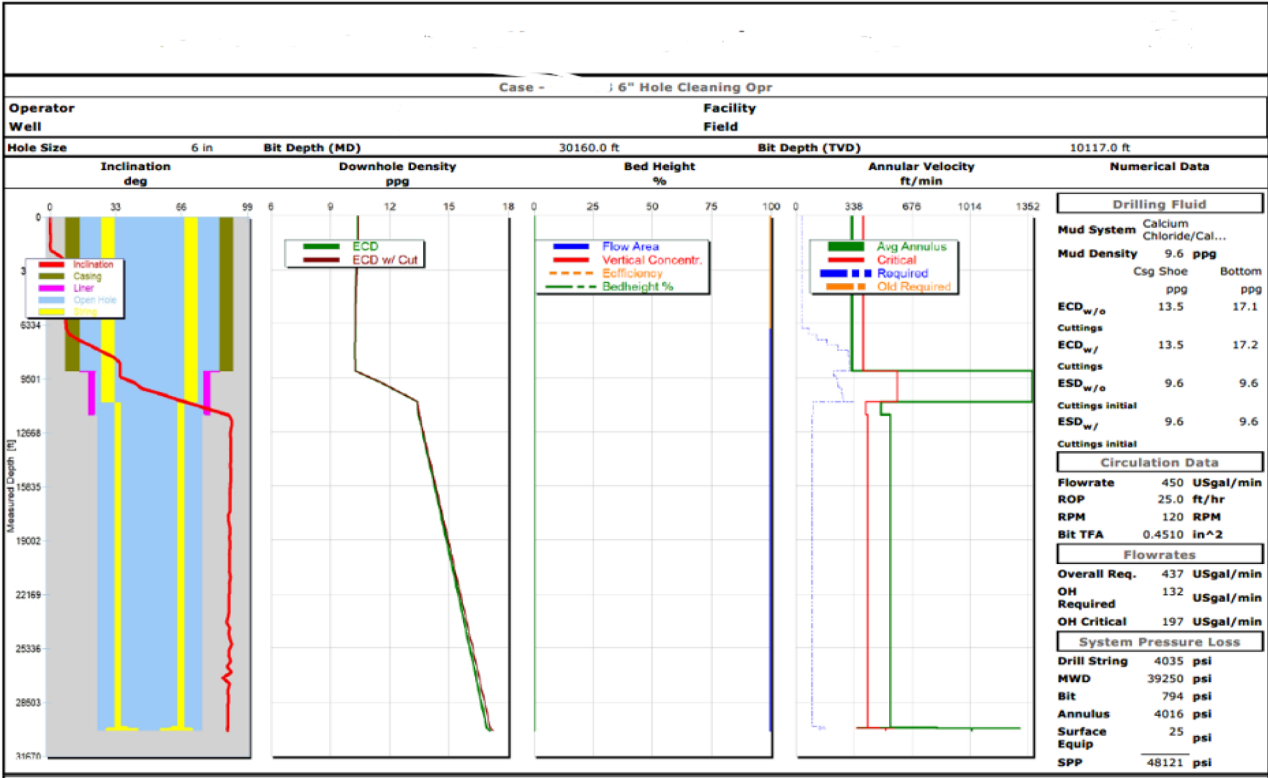


Figure 9—ECD, Hole cleaning during drilling

Temperature profile

This sets a new global benchmark for WBM reach, extending ~30% beyond traditional WBM limits and matching distances previously possible only with oil-based mud (OBM). Second, despite encountering extreme bottomhole temperatures (~149 °C / 300 °F), the operation maintained thermal control by deploying dual mud chillers to cool the fluid and a high-temperature lubricant to reduce friction. This proactive temperature management kept the circulating mud temperatures in check and preserved fluid integrity throughout the 6- day drilling period. Third, the WBM exhibited exceptional stability under these harsh conditions - rheological properties remained within design limits, no barite sag or gelation occurred, and high-temperature fluid loss stayed low. [Figure 5](#)- Showing the temperature profile while drilling

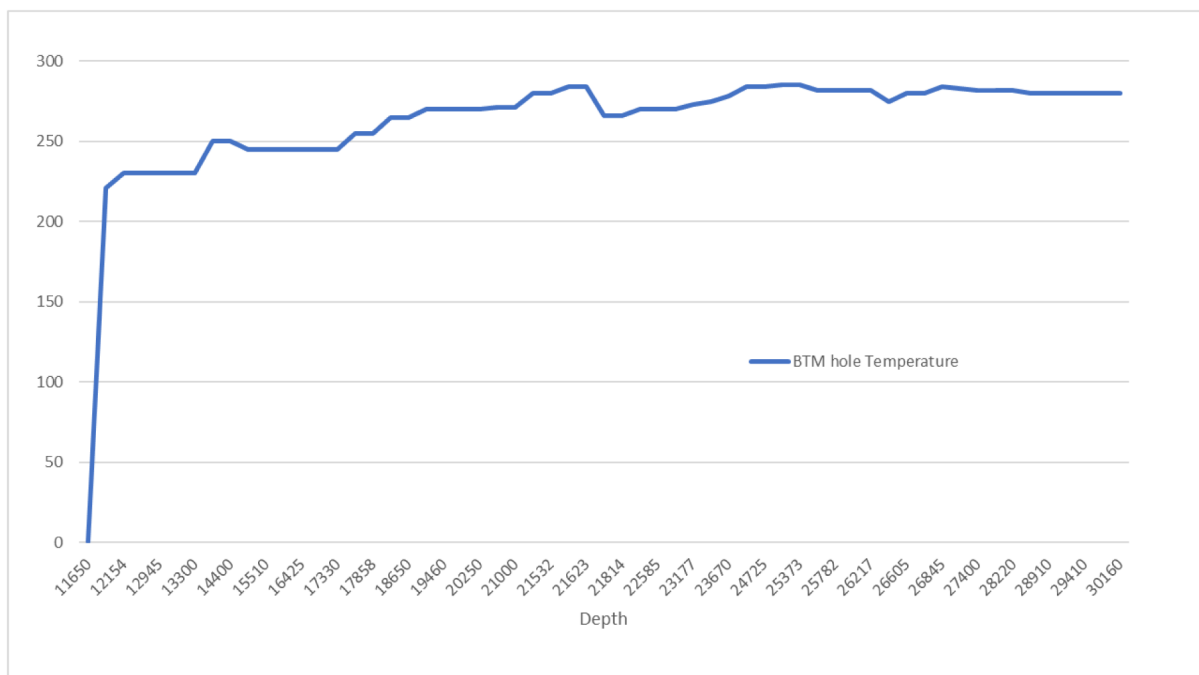


Figure 10—Showing the temperature profile while drilling

Environmental and Economic Benefits Realized

Using an environmentally friendly WBM instead of RDF NAF had clear advantages. The operation avoided the use of thousands of barrels of oil-based fluids and associated cuttings waste. This aligns with the new vision and the industry's drive for more sustainable drilling practices. Additionally, substantial cost savings (~ 48 % per well) were realized by using a WBM, without sacrificing performance. Over a multi-well campaign, this translates to tens of millions in savings and a much lower ecological footprint for the project.

Figure 11- WBM cost savings per well

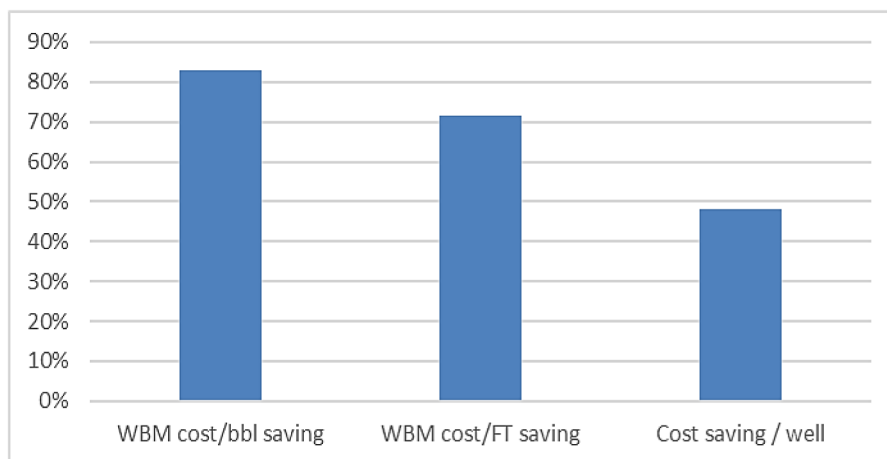


Figure 11—WBM cost savings per well

Conclusions

In summary, the project successfully overcame significant technical challenges —such as hydraulic optimization, high torque and friction, fluid contamination, and maintaining fluid performance at elevated temperatures —while employing environmentally responsible fluids and advanced management strategies. This operation proves the technical viability of WBM for ultra-ERD applications, challenging the long-

held belief that only more complex systems could achieve such objectives. Historically, WBM has been confined to horizontal sections under 14,000 feet; however, this project extended that limit by over 30%, redefining the possibilities in Maximum Reservoir Contact (MRC) well architecture. It establishes a new industry benchmark for environmentally conscious drilling, offering a sustainable, cost-efficient alternative for unlocking complex reservoirs

Acknowledgments

Nomenclature

ECD	=	Equivalent circulating density
WBM	=	Water-Based Mud
KPI	=	Key performance indicator
SOP	=	Standard Operating Procedures
MD	=	Measured depth
TVD	=	True vertical depth
NDF Reservoir	=	Non-Damaging Water-Based Mud

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